

Ontological Incompleteness of Closed Theories: The Necessity of a Latent Domain

1. Introduction

Modern foundational theories frequently aspire to *self-sufficiency*. Whether in physics, philosophy of science, or metaphysics, there is a recurring ideal: a theory should explain reality entirely in its own terms, without recourse to external principles, latent structures, or non-observable domains. In its strongest form, this ideal gives rise to what may be called **self-sustaining theories**—frameworks that claim to account for both phenomena and their conditions of intelligibility using only entities defined within an observable or operational domain.

This aspiration has shaped large portions of twentieth- and twenty-first-century foundational discourse. In quantum mechanics, it appears as resistance to any ontology beyond measurement outcomes or Hilbert-space formalism. In philosophy, it manifests as strict physicalism or anti-metaphysical naturalism. In each case, appeals to latent structure are often dismissed as unnecessary, unscientific, or "merely metaphysical."

This paper argues that such dismissal is misplaced. The central claim is not that theories *should* introduce additional ontological layers, but that **they already do so implicitly**. Any theory formulated exclusively within an observable domain—denoted here as $(\mathcal{N}_0)$ —necessarily presupposes a latent structural domain, $(\mathcal{N}_1)$, in order to account for its own realizability, stability, and selection of outcomes. When this presupposition is denied, it does not disappear; it is merely concealed.

We advance three tightly connected theses:

1. No theory confined to $(\mathcal{N}_0)$ can be ontologically complete.
2. Every purportedly self-sustaining theory contains a disguised circularity of the form $(X \rightarrow X)$, where the theory legitimizes its own conditions of validity.
3. What is commonly labeled "metaphysical excess" is, in fact, the unavoidable structural remainder left by the incompleteness of $(\mathcal{N}_0)$.

The contribution of this paper is not the introduction of new mathematical tools, but the **explicit isolation of a structural distinction that is already operative** across foundational theories. By formalizing the roles of $(\mathcal{N}_0)$, $(\mathcal{N}_1)$, and the projection between them, we show that ontological closure is not merely difficult to achieve—it is impossible in principle.

2. Observable Closure and the Myth of Self-Sufficiency

A theory is commonly said to be *closed* when it satisfies two conditions: (i) all of its entities are defined internally, and (ii) all of its explanatory moves refer only to those entities. In practice, this notion of closure is usually restricted to what can be observed, measured, or operationally accessed. We refer to this domain as the **observable domain**, $(\mathcal{N}_0)$.

Observable closure, however, admits several distinct meanings that are often conflated:

- **Empirical closure**: all predictions concern observable outcomes.
- **Formal closure**: the theory is mathematically self-contained.
- **Ontological closure**: the theory accounts for the existence and structure of its own primitives.

While empirical and formal closure are routinely achieved, ontological closure is not. Yet it is ontological closure that self-sustaining theories implicitly claim.

The problem becomes apparent when a theory attempts to explain not merely *what happens*, but *why only certain possibilities are realized*. Selection, stability, and necessity cannot be reduced to raw observation. They presuppose a space of alternatives—most of which are *not* observed. A theory confined to $(\mathcal{N}_0)$ can describe realized outcomes, but it cannot, without contradiction, account for the structure of unrealized possibilities.

This gap is typically bridged through conceptual maneuvers that preserve the appearance of closure: probabilistic postulates, axiomatic constraints, or interpretational rules treated as primitive. These maneuvers do not eliminate the need for a latent domain; they merely rename it.

Thus, the myth of self-sufficiency arises not from logical consistency, but from **category compression**: conditions of possibility are forced into the same domain as realized phenomena. The result is a theory that appears closed, while remaining ontologically incomplete.

3. Formal Definitions: $(\mathcal{N}_0)$, $(\mathcal{N}_1)$, and Projection

To make this claim precise, we introduce a minimal structural distinction.

- **Observable domain** $(\mathcal{N}_0)$: the set of realized states, events, or outcomes accessible to observation, measurement, or operational definition within a theory.
- **Latent structural domain** $(\mathcal{N}_1)$: the set of structurally admissible possibilities that are not directly observable, but are required to define necessity, stability, or selection within $(\mathcal{N}_0)$.

The relation between these domains is not symmetric. $(\mathcal{N}_0)$ depends on $(\mathcal{N}_1)$ for its intelligibility, but not vice versa. This dependence is formalized by a **projection map**:

$$[\pi : \mathcal{N}_1 \rightarrow \mathcal{N}_0,]$$

which maps latent possibilities to realized outcomes under specific constraints.

Crucially, $(\mathcal{N}_1)$ is not an additional physical space, nor does it introduce new empirical entities. It is a **structural domain**, defined only by its necessity for explaining why $(\mathcal{N}_0)$ has the form it does rather than another.

Any attempt to eliminate $(\mathcal{N}_1)$ requires that the projection (π) be identity, collapsing possibility into actuality. This move renders selection unintelligible and forces the theory into circular justification: outcomes are necessary because they occur, and they occur because they are necessary.

The distinction between $(\mathcal{N}_1)$ and $(\mathcal{N}_0)$ therefore does not add metaphysics; it **makes explicit what closure already presupposes**. Subsequent sections will show that denying this distinction leads inevitably to hidden circularity and ontological incompleteness.

4. The Ontological Incompleteness of $\mathcal{N}_0$

We now formalize a claim that is usually avoided rather than refuted:

■ *The observable domain $\mathcal{N}_0$ cannot be ontologically complete.*

This is not a metaphysical preference. It is a structural consequence of how explanation, determination, and realization function in any physical theory.

4.1 What $\mathcal{N}_0$ Contains — and What It Cannot

By definition, $\mathcal{N}_0$ is the domain of:

- observable states,
- measurement outcomes,
- empirical records,
- operationally accessible quantities.

Any theory formulated *entirely* within $\mathcal{N}_0$ must explain all of its elements using only relations among those same elements.

This immediately creates a problem: **explanation collapses into description**.

If every realized state is explained only by reference to other realized states, then no account can be given of:

- why *this* outcome occurred rather than another,
- why certain states are admissible at all,
- why stability appears in preferred bases.

Any such explanation requires reference to **conditions of realizability**, not merely to realized facts.

Those conditions cannot themselves belong to $\mathcal{N}_0$ without circularity.

4.2 Ontological Circularity as a Necessary Failure Mode

Suppose a theory claims to be fully self-sustaining within $\mathcal{N}_0$.

Then, for every element $x \in \mathcal{N}_0$, its explanation must be of the form:

x is explained by a set of elements $x_1, x_2, \dots, x_n$, all belonging to $\mathcal{N}_0$.

But if any x_i is itself unexplained except by other elements of $\mathcal{N}_0$, the structure necessarily closes into a loop.

Formally, such theories contain a disguised mapping:

$$\mathcal{N}_0 \rightarrow \mathcal{N}_0$$

with no external grounding.

This is the defining signature of a **self-sustaining (autosustentable) theory**.

The problem is not logical inconsistency; the problem is **ontological vacuity**. Nothing in the theory explains *why* the structure exists or *why* certain states are selected.

Circularity here is not an accident. It is **necessary** if $\mathcal{N}_1$ is denied.

4.3 Ontological Incompleteness Theorem for $\mathcal{N}_0$

Ontological Incompleteness Theorem

Let $\mathcal{N}_0$ be a theory-defined domain containing only realized, observable, or operationally accessible states, and assume that the theory formulated over $\mathcal{N}_0$ is claimed to be self-sustaining, in the sense that it explains both its outcomes and the conditions of their realization using only elements internal to $\mathcal{N}_0$.

Then:

No such theory can be ontologically complete.

More precisely:

Any theory restricted to $\mathcal{N}_0$ that accounts for selection, stability, or necessity of its outcomes necessarily presupposes a latent structural domain $\mathcal{N}_1$ and a non-identity projection from $\mathcal{N}_1$ to $\mathcal{N}_0$. If this latent domain is denied explicitly, it reappears implicitly as circular explanation of the form X explains X.

Equivalently:

Ontological closure within $\mathcal{N}_0$ is impossible without concealed circularity.

Corollary.

Every self-sustaining theory contains an implicit $\mathcal{N}_1$, whether acknowledged or not. Denial of $\mathcal{N}_1$ does not eliminate it; it only obscures its role.

Theorem (Ontological Incompleteness of $\mathcal{N}_0$)

Any theory restricted to the observable domain $\mathcal{N}_0$ is either

1. ontologically incomplete, or
2. circular (self-sustaining), or
3. both.

There is no third option.

We can now state the result informally but precisely.

Ontological Incompleteness Theorem ($\mathcal{N}_0$):

Any theory restricted to the observable domain $\mathcal{N}_0$ is either:

- 1. ontologically incomplete, or*
- 2. circular (self-sustaining), or*
- 3. both.*

There is no third option.

Attempts to avoid this conclusion typically proceed by re-labeling $\mathcal{N}_1$ as:

- “law of nature”,
- “probability postulate”,
- “environment”,
- “branching structure”,
- “measurement axiom”.

But renaming does not remove ontological function.

If something **selects**, **constrains**, or **projects**, it is not in $\mathcal{N}_0$.

Calling it “just formalism” does not change its role.

5. Hidden $\mathcal{N}_1$ in Standard Interpretations

We now show that **every major interpretation of quantum mechanics already assumes an $\mathcal{N}_1$** , even when explicitly denying it.

The disagreement is not about existence — only about admission.

5.1 Copenhagen-Type Interpretations

Copenhagen-style views assert that:

- measurement outcomes are not determined prior to measurement,
- values are created in the act of observation.

However, the *measurement context* itself determines:

- which observables are admissible,
- which eigenstates are stable,
- which outcomes can occur.

That contextual structure is not itself observable as an outcome.

It therefore belongs to a latent domain functionally equivalent to $\mathcal{N}_1$.

Denying its ontological status while relying on its selective role is incoherent.

5.2 Many-Worlds Interpretations

Many-worlds attempts to avoid collapse by asserting that all outcomes occur.

But this immediately raises two questions:

1. Why do branches have a preferred basis?
2. Why are observers confined to quasi-classical branches?

The answer is always structural: decoherence, stability, environment-induced selection.

Those structures determine **which possibilities are realized as worlds**.

They are not themselves worlds.

Thus, even here, realization depends on a latent selection structure — an $\mathcal{N}_1$ — merely distributed rather than denied.

5.3 Objective Collapse Models

Objective collapse theories introduce explicit stochastic dynamics.

But the noise terms, collapse rates, or localization operators:

- are not observable outcomes,
- are not states of the system,
- determine what becomes real.

They therefore occupy the ontological role of $\mathcal{N}_1$, even when framed as “physical randomness”.

Randomness does not eliminate ontology; it *is* an ontological commitment.

5.4 Decoherence-Only Accounts

Pure decoherence accounts insist that no collapse occurs.

Yet decoherence singles out:

- pointer states,
- stable subspaces,
- dynamically robust structures.

The environment here acts as a **boundary operator** selecting admissible states.

Again, this selecting structure is not itself an outcome.

It is a latent constraint space — an implicit $\mathcal{N}_1$.

5.5 Summary Table (Conceptual)

INTERPRETATION	EXPLICIT $\mathcal{N}_1$	IMPLICIT $\mathcal{N}_1$
Copenhagen	No	Measurement context
Many-Worlds	No	Branching / decoherence
Objective Collapse	Yes (stochastic)	Noise / collapse operators
Decoherence-only	No	Environment-induced structure

The conclusion is unavoidable:

*Standard interpretations do not avoid $\mathcal{N}_1$.
They merely leave it unnamed.*

A familiar historical example illustrates the necessity of a latent domain. In classical thermodynamics, macroscopic variables such as temperature and pressure belong to the observable domain. Yet their stability and necessity cannot be explained without reference to a latent space of microstates. The introduction of phase space did not add empirical entities; it made explicit the structural domain required for macroscopic regularities to be intelligible.

The relation between microstates and macrostates plays, at a structural level, the same role as the $N_I \rightarrow N_o$ projection discussed here.

Similarly, in quantum mechanics, the existence of a preferred basis cannot be explained solely at the level of observed outcomes. The stability of pointer states presupposes a space of admissible possibilities from which only a subset is realized under environmental constraints.

6. Formal Characterization of $N_1 \rightarrow N_0$ as Projection

We now make explicit the structural relation that has remained implicit throughout the previous sections: the transition from latent possibility to realized outcome.

6.1 Projection as a Structural, Not Epistemic, Map

The relation between the latent structural domain (N_1) and the observable domain (N_0) is best understood as a **projection**, not as an inference, update of knowledge, or act of observation.

A projection is characterized by three minimal properties:

1. **Asymmetry**: N_0 depends on N_1 for its intelligibility, but N_1 does not depend on N_0 .
2. **Non-invertibility**: multiple latent possibilities in N_1 can map to the same realized state in N_0 .
3. **Constraint sensitivity**: the mapping depends on boundary conditions and structural constraints.

In schematic form, this relation is written simply as:

$$\mathcal{N}_1 \rightarrow \mathcal{N}_0$$

This arrow does not denote temporal evolution, causal interaction, or epistemic update. It denotes **ontological reduction under constraint**.

The projection $\pi: N_1 \rightarrow N_0$ can also be understood as the formal realization of a cardinal collapse from $\aleph_1$ to $\aleph_0$, as previously described in the Axiomatic Necessity Theorem (Bresciano, 2026, v1.2, DOI: 10.5281/zenodo.18280834). In this sense, projection does not introduce a new mechanism, but provides a structural expression of necessity reduction already implicit in the theory.

6.2 Why Projection Cannot Be Identity

If the projection from N_1 to N_0 were identity, then every possibility would be realized, and no selection would occur. In such a scenario:

- there would be no stable states,
- no preferred bases,
- no notion of necessity or robustness.

The existence of structured, repeatable, and stable outcomes therefore already implies that the projection is non-trivial.

Any theory that denies projection while retaining selection is internally inconsistent. Selection without projection is indistinguishable from magic.

Consider a simple schematic case. Let N_i contain a continuum of structurally admissible states compatible with a given operator or constraint. Boundary conditions restrict this space, and the projection π maps an entire equivalence class of latent possibilities onto a single stable outcome in N_o .

From the perspective of N_o , this appears as “collapse.” From the perspective of structure, it is selection under constraint.

6.3 Projection Without Metaphysical Inflation

Crucially, introducing projection does not add entities to the theory. It adds **explicit structure**.

Projection answers a question that every theory already presupposes but rarely states:

Why do only certain possibilities become actual?

By formalizing $\mathcal{N}_1 \rightarrow \mathcal{N}_0$ as projection, we avoid appeals to:

- ontological randomness,
- observer-induced creation,
- unexplained axiomatic collapse.

The projection is not a new physical process. It is the **structural interface** between possibility and realization.

7. Consequences for Quantum Foundations and Metaphysics

Making the $\mathcal{N}_1 \rightarrow \mathcal{N}_0$ projection explicit has immediate consequences for both quantum foundations and metaphysical interpretation.

7.1 Dissolution of the Measurement Pseudo-Problem

Once projection is acknowledged as a structural necessity, the so-called measurement problem changes status.

The problem is not that quantum mechanics lacks a mechanism of collapse. The problem is that collapse has been misinterpreted as:

- ontological creation,
- stochastic miracle,
- observer-dependent event.

Under the projection framework, measurement is simply the **realization of a constrained possibility**.

Nothing is created. Nothing mysterious occurs. What changes is domain: from N_1 to N_0 .

7.2 Why Metaphysics Re-enters Through the Back Door

Attempts to eliminate metaphysics by restricting discourse to N_0 invariably fail. They merely displace metaphysical commitment into:

- axioms labeled “fundamental”,
- probabilities treated as primitive,
- laws treated as self-explanatory.

These moves do not remove N_1 ; they hide it.

The projection framework restores conceptual honesty by making explicit what was already assumed.

7.3 Structural Minimalism as a Constraint, Not a License

Importantly, acknowledging N_1 does not grant license to speculative ontology. On the contrary, it imposes discipline.

N_1 is not:

- a realm of hidden variables,
- a second physical universe,

- a mental or conscious domain.

It is only what is **strictly required** for N_0 to be intelligible.

This is methodological minimalism applied at the ontological level.

7.4 Reframing the Status of Foundational Debates

Many long-standing disputes in quantum foundations arise from competing attempts to either:

- deny N_1 , or
- overpopulate it.

The present framework rejects both extremes.

By recognizing N_1 as unavoidable but minimal, we obtain a stable interpretational stance in which:

- realism does not require determinism,
- indeterminism does not require ontological randomness,
- explanation does not require observers.

7.5 Summary

The distinction between N_1 and N_0 , together with their projection relation, does not solve quantum mechanics by adding ingredients. It clarifies quantum mechanics by removing confusion.

What has long been treated as a metaphysical embarrassment is, in fact, a structural necessity.

Once this is seen, the insistence on self-sustaining theories appears not as rigor, but as avoidance.

Ontology does not disappear when it is ignored. It merely becomes implicit—and unexamined.

The present framework is naturally aligned with recent discussions in structural realism, particularly ontic structural realism, insofar as it assigns primacy to relational and spectral structure rather than to individual events. However, it departs from standard formulations by making explicit the necessity of a latent structural domain rather than restricting ontology to observable relations alone.

Likewise, while decoherence theory successfully explains the dynamical suppression of interference, it typically treats the emergence of classical outcomes as an effective phenomenon. The $N_1 \rightarrow N_0$ projection framework complements decoherence by providing an ontological reading of why such effective structures are stable and necessary, rather than merely approximate.

8. Ontological Incompleteness of N_0

We are now in a position to state explicitly what has remained implicit throughout the previous analysis.

8.1 Formal Statement

Theorem (Ontological Incompleteness of N_0).

No theory formulated exclusively within the observable domain N_0 can be ontologically complete. Any such theory necessarily presupposes a latent structural domain N_1 that is not definable entirely within N_0 .

8.2 Clarification of Terms

Ontological completeness means that a theory accounts not only for observed outcomes, but also for:

- why those outcomes are possible,
- why they are stable,
- why they are selected rather than alternatives.

A theory confined to N_0 may describe what occurs, but cannot, without contradiction, explain why only those occurrences are admissible.

8.3 Proof Sketch

Assume, for contradiction, that a theory T formulated entirely within N_0 is ontologically complete.

Because T is complete, it must explain:

- the existence of its own primitives,
- the stability of its states,

- the selection of outcomes among alternatives.

However, explanation of selection requires reference to unrealized possibilities. These possibilities cannot belong to N_0 , since N_0 contains only realized states.

Therefore, either:

1. unrealized possibilities are denied — in which case selection becomes unintelligible, or
2. unrealized possibilities are implicitly assumed — in which case a latent domain exists.

In both cases, N_0 alone is insufficient.

Thus, ontological completeness of N_0 leads to contradiction.

Therefore, N_0 is ontologically incomplete.

8.4 Consequence

The incompleteness of N_0 is not empirical, mathematical, or epistemic. It is structural.

Attempts to avoid this conclusion do not eliminate N_1 ; they merely encode it implicitly through axioms, primitives, or interpretational rules.

9. Where N_1 Is Hidden in Standard Interpretations

The necessity of a latent domain is not a peculiarity of the present framework. On the contrary, all major interpretations of quantum mechanics already rely on an implicit N_1 . They differ only in how openly this reliance is acknowledged.

9.1 Copenhagen-Type Interpretations

Copenhagen interpretations restrict ontology to measurement outcomes and classical descriptions.

However, they presuppose:

- a space of possible outcomes prior to measurement,
- a rule selecting one outcome,
- a classical-quantum cut.

This pre-measurement possibility space functions as N_1 , even though it is denied ontological status.

N_1 is present, but declared unspeakable.

9.2 Many-Worlds Interpretations

In many-worlds approaches, all possibilities are realized.

Here, N_1 is identified with the universal wavefunction itself.

Projection is replaced by branching, but the latent domain remains: the space of all branches exists prior to any observer-relative realization.

The disagreement is therefore not about whether N_1 exists, but about whether projection is allowed.

9.3 Bohmian Mechanics

Bohmian mechanics introduces particle positions guided by a wavefunction.

The guiding wave defines a space of admissible trajectories that exceeds what is observed.

This guiding structure is precisely a latent domain: N_1 is explicit and dynamical.

Bohmian mechanics is honest about N_1 , but pays the price of additional structure.

9.4 Decoherence-Based Approaches

Decoherence explains the suppression of interference and the emergence of preferred bases.

However, decoherence alone does not explain why a single outcome is realized.

The space of decohered but unrealized alternatives constitutes an implicit N_1 .

Decoherence describes constraint, not projection.

9.5 QBism and Participatory Interpretations

QBism relocates quantum states to agents' beliefs.

Nevertheless, the space of possible belief updates and admissible experiences still precedes any particular outcome.

N_1 is relocated to the normative structure of rational expectation.

Ontology is avoided rhetorically, but structure remains.

9.6 Summary

Across interpretations, the pattern is invariant:

- N_1 is always present,
- projection is always required,
- denial of N_1 produces hidden circularity.

Interpretational disagreement concerns how N_1 is labeled, not whether it exists.

10. Circularity, Gödel, and Why Self-Sustaining Theories Must Close on Themselves

The accusation most frequently directed at self-referential or circular theories is that they are logically defective. Circularity is commonly associated with vicious regress, tautology, or explanatory failure. This section argues the opposite: **for a theory that aspires to be self-sustaining, circularity is not a flaw but a structural necessity.**

The error lies not in recognizing circularity, but in misunderstanding its role.

10.1 The Ideal of the Self-Sustaining Theory

A self-sustaining theory is one that claims to satisfy the following condition:

- all entities it refers to,
- all principles it employs,
- and all criteria of validity it invokes

are defined internally, without appeal to anything external.

In other words, the theory claims to justify itself entirely within its own domain.

This ideal appears across disciplines:

- in physics, as theories that explain both phenomena and laws using only internal structure,
- in philosophy, as naturalistic or anti-metaphysical frameworks,
- in logic, as formal systems aspiring to semantic closure.

What is rarely acknowledged is that **this ideal already implies circularity**.

10.2 The Inevitability of $X \rightarrow X$

Consider any theory T that claims ontological closure within N_0 .

To be self-sustaining, T must explain:

- why its primitives exist,
- why its rules are valid,
- why its outcomes are admissible.

But these explanations can only appeal to elements already defined by T .

Therefore, at some level, the theory must assert:

- its primitives are valid because the theory says so,
- its rules hold because they are rules of the theory,
- its outcomes are necessary because they occur within the theory.

This produces a closed explanatory loop of the form:

X is justified by X .

The loop may be deeply disguised—spread across axioms, probability postulates, or interpretational rules—but it is structurally unavoidable.

A self-sustaining theory **cannot escape $X \rightarrow X$** without ceasing to be self-sustaining.

10.3 Gödel as a Structural Precedent, Not a Limitation

Gödel's incompleteness theorems are often interpreted narrowly as results about arithmetic or formal logic. Their deeper significance is structural.

Gödel showed that:

- any sufficiently expressive formal system cannot prove its own consistency from within,
- statements about the system as a whole necessarily transcend the system.

This is not a defect of particular axioms. It is a consequence of closure.

In the present framework, N_0 plays a role analogous to a formal system, while N_1 corresponds to the meta-structural domain required to make sense of it.

Ontological incompleteness is therefore not an accident. It is the ontological analogue of Gödelian incompleteness.

10.4 Why Circularity Is Necessary, Not Pathological

The crucial distinction is between:

- **vicious circularity**, which explains nothing new, and
- **structural circularity**, which stabilizes a closed system.

Self-sustaining theories require structural circularity in order to terminate explanation. Without it, they collapse into infinite regress.

The mistake of many foundational programs is to:

- deny circularity while relying on it implicitly, or
- treat circularity as failure rather than as closure.

Once circularity is acknowledged explicitly, its role becomes clear:

- it marks the boundary of the theory,
- it signals where projection must occur,
- it reveals where N_1 has been suppressed.

10.5 Circularity and the Disguise of N_1

When a theory denies the existence of a latent domain, circularity becomes its only remaining option.

Selection is explained by axioms.

Axioms are justified by necessity.

Necessity is inferred from outcomes.

Thus, the projection from N_1 to N_0 is replaced by a closed loop within N_0 itself.

This is why self-sustaining theories often appear formally elegant yet conceptually opaque. They have compressed N_1 into N_0 by force.

The result is not metaphysical purity, but **ontological amnesia**.

10.6 Consequence for Foundational Discourse

The choice is not between circularity and non-circularity.

The real choice is between:

- explicit acknowledgment of a minimal latent domain with projection, or
- implicit circularity masquerading as completeness.

The former preserves clarity.

The latter preserves ideology.

Once this is understood, accusations of “metaphysics” lose their force. What is being rejected is not excess ontology, but the exposure of what closure already requires.

10.7 Summary

Self-sustaining theories must close on themselves. They have no alternative.

Gödel did not show that closure is impossible. He showed that closure has a price.

In ontological terms, that price is the necessity of a latent structural domain—or, failing that, an unavoidable circularity disguised as explanation.

The next section draws the final implications of this result for realism, interpretation, and the future of foundational theory.

11. Conclusion

This paper has argued for a simple but non-negotiable claim: **no theory confined to an observable or operational domain can be ontologically complete**. Any framework that aspires to be self-sustaining must either acknowledge a latent structural domain or accept a concealed circularity in its own explanatory core.

The distinction between an observable domain (N_0) and a latent structural domain (N_1) does not introduce new physics, new entities, or speculative metaphysics. It merely makes explicit what closure already presupposes. Selection, stability, necessity, and realization cannot be accounted for within N_0 alone without collapsing explanation into self-justification.

Attempts to deny this distinction have followed a consistent pattern. Latent structure is rejected at the ontological level, only to reappear as primitive axioms, unexplained probabilities, or interpretational rules treated as irreducible. In each case, N_1 is not eliminated but hidden.

By formalizing the relation between N_1 and N_0 as projection, we have shown that:

- circularity is unavoidable in self-sustaining theories,
- ontological incompleteness is structural rather than accidental,
- and accusations of “metaphysical excess” often mask unacknowledged dependence on latent structure.

This reframes several long-standing foundational debates. The question is not whether metaphysics should be excluded, but whether it should be **explicit and minimal**, or **implicit and uncontrolled**. Structural minimalism favors the former.

The framework presented here does not compete with existing interpretations of quantum mechanics or other foundational theories. Instead, it provides a diagnostic lens through which their hidden commitments can be made visible. Where a theory claims closure, one may now ask a precise question: *where has projection been concealed?*

In this sense, the contribution of this work is not additive but clarificatory. It does not propose a new ontology. It restores conceptual honesty by showing that what is often dismissed as metaphysics is, in fact, the unavoidable remainder of closure itself.

Ontological completeness is not something a theory can achieve.
But ontological clarity is.

And clarity, in foundational theory, is progress.

Ontological completeness is impossible. But ontological clarity — the explicit acknowledgment of N_i — is not only possible; it is the first step toward structural honesty in foundational theory.

Appendix A — Intuitive Overview of the N_1 / N_0 Framework

A.1 Purpose and Scope

This appendix provides an intuitive, non-technical overview of the distinction between the latent structural domain (N_1) and the observable domain (N_0) introduced in the main text. It is intended for readers outside quantum foundations or philosophy of physics, as well as for those encountering the framework for the first time.

No new claims, results, or formal structures are introduced here. The purpose is solely explanatory.

A.2 Observable Outcomes and the Problem of Selection

Most physical theories describe what is observed: measurement outcomes, recorded events, or stable macroscopic states. These constitute what we call the observable domain, N_0 .

However, observation alone does not explain **why** particular outcomes occur rather than others. The existence of stable, repeatable results presupposes a space of alternatives—most of which are never observed.

Any account that includes notions such as:

- stability,
- robustness,
- necessity,
- or selection,

implicitly refers to possibilities that were available but not realized.

This simple fact already exceeds what can be described using only realized outcomes.

A.3 Latent Possibility as a Structural, Not Speculative, Domain

The latent domain, N_1 , is introduced to account for these unobserved possibilities. It is important to emphasize what N_1 is **not**:

- It is not a hidden physical space.
- It is not a second universe.
- It is not a mental or observer-dependent realm.

N_1 is a **structural domain**: the set of all possibilities that are compatible with the formal structure of the theory before constraints are applied.

A familiar analogy comes from thermodynamics. Macroscopic variables such as temperature or pressure belong to the observable domain. Their stability and necessity, however, cannot be explained without reference to a space of microstates. The introduction of phase space did not add metaphysical excess; it made explicit what was already required.

N_1 plays an analogous role.

A.4 Projection: From Possibility to Realization

The relation between N_1 and N_0 is expressed by a projection:

$$[\pi : \mathcal{N}_1 \rightarrow \mathcal{N}_0,]$$

This projection represents the transition from structural possibility to realized outcome under specific constraints.

Crucially, projection is:

- **non-invertible**: many possibilities can correspond to the same observed result,
- **asymmetric**: N_0 depends on N_1 for its intelligibility, but not vice versa,
- **constraint-dependent**: boundary conditions determine which possibilities are realized.

From within N_0 , this process may appear abrupt or mysterious. From the structural perspective, it is simply selection under constraint.

A.5 Why Projection Is Not Optional

If no distinction were drawn between N_1 and N_0 , every possibility would be identical to its realization. In such a scenario:

- there would be no meaningful notion of selection,
- no explanation of stability,
- no distinction between necessity and accident.

The very existence of structured outcomes therefore already implies a latent domain.

Attempts to deny N_1 do not eliminate it; they merely conceal it behind axioms, probabilities treated as primitive, or unexplained selection rules.

A.6 Implications for Quantum Measurement

In quantum mechanics, measurement is often described as a problem because outcomes appear to be “created” during observation.

Within the N_1/N_0 framework, no creation occurs. Measurement corresponds to the projection of a constrained set of quantum possibilities into a realized outcome.

What changes is not reality itself, but **the domain in which it is described**.

This perspective reframes collapse as a structural transition rather than a physical anomaly.

A.7 Minimalism and Conceptual Honesty

Introducing N_1 does not expand ontology beyond necessity. On the contrary, it prevents uncontrolled metaphysical inflation by isolating precisely what is required and nothing more.

The framework does not claim to solve all foundational questions. Its claim is narrower:

A theory confined entirely to observable outcomes cannot account for its own conditions of realization.

Recognizing this fact is not metaphysical extravagance; it is conceptual honesty.

A.8 Closing Remark

The N_1/N_0 distinction is not an alternative interpretation competing with existing ones. It is a structural clarification that reveals what many interpretations already assume implicitly.

By making this assumption explicit, the framework aims to replace hidden circularity with transparent structure—and to do so with the minimum ontological commitment required for intelligibility.